Business Case Study

Business Case: Scaler Clustering

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Colab Link: https://drive.google.com/file/d/1f7Knb8krsRWKah-th_VxZiTm8Nw8TsLk/view?usp=drive_link

Problem Statement

Scaler is an online tech-versity offering intensive computer science & Data Science courses through live classes delivered by tech leaders and subject matter experts. The meticulously structured program enhances the skills of software professionals by offering a modern curriculum with exposure to the latest technologies. It is a product by InterviewBit.
You are working as a data scientist with the analytics vertical of Scaler, focused on profiling the best companies and job positions to work for from the Scaler database. You are provided with the information for a segment of learners and tasked to cluster them on the basis of their job profile, company, and other features. Ideally, these clusters should have similar characteristics.

Dataset
Dataset Link: ***scaler clustering***
Data Dictionary:

- 'Unnamed 0' - Index of the dataset
- Email_hash - Anonymised Personal Identifiable Information (PII)
- Company_hash - This represents an anonymized identifier for the company, which is the current employer of the learner.
- orgyear - Employment start date
- CTC - Current CTC
- Job_position - Job profile in the company
- CTC_updated_year - Year in which CTC got updated (Yearly increments, Promotions)

Concept Used:

- Manual Clustering
- Unsupervised Clustering - K- means, Hierarchical Clustering

Key insights :

1. Data quality

- The raw file has 205,843 rows but only **153,443 unique learners**. After removing 34 exact duplicates and keeping the latest record per learner, about 25% of rows were repeat updates.
- CTC is extremely skewed: median ₹9.5 lakh, mean ₹22.7 lakh, 99th percentile ₹126 lakh and 99.9th percentile ₹20 crore. On the log scale

the skew drops from 15.97 to −0.24, so log or median is the right way to analyse salary.

- The IQR rule on log CTC accepts ₹1.01 lakh to ₹92.38 lakh and removes **7,146 records (4.66%)**. There are about twice as many too-low values (4,761) as too-high (2,385), which points to unit errors such as monthly pay or values entered in thousands.
- The role field is weak: **25.5% missing** and **1,016 distinct free-text values**. Company has 37,299 distinct names, and the top 15 cover only 21% of records. Segmentation therefore rests on pay and experience, not on role or employer.
- The cleaning handled 1,301 impossible joining years, 85 purely numeric company names and 1,179 learners with imputed experience.

2. EDA relationships

- Experience and log CTC correlate at only **0.34** (Spearman), so experience explains part of pay but role, employer and skills matter too.
- Experience and update year correlate at −0.26: experienced learners tend to have older salary records, and recent updates come mostly from early-career learners.

3. Manual flags

Flag	Coverage	Above mean (1)	Below mean (3)	Median CTC (1 vs 3)	Median experience (1 vs 3)
Designation	38.9%	15.9%	22.9%	₹15L vs ₹6.4L	5 vs 4 yrs
Class	62.3%	24.8%	37.4%	₹18L vs ₹7L	6 vs 4 yrs
Tier	78.0%	29.8%	48.2%	₹19L vs ₹7.1L	7 vs 5 yrs

- Coverage rises from Designation to Tier, but the like-for-like comparison weakens. The experience gap between flag 1 and flag 3 widens from 1 year to 2 years, so Class and Tier increasingly measure seniority, not fair pay.
- Outlier removal reduced the skew bias: below-mean learners now outnumber above-mean by 1.44 to 1.61 per one, down from about 1.9 to 2.9 before cleaning.
- **30,354 learners (20.7%)** are below peer mean on both Designation and Tier.
- Data/ML roles (4,418 records) have a median of ₹12L against ₹10L for all roles, a premium of roughly 20%.
- Top company medians range ₹30 to 39 lakh, but most rest on only 20 to 41 records. **dgzo** (n=148, median ₹34.25L) is the most reliable.

4. Clustering

- The clustering tendency is strong: Hopkins **0.928** on unique rows.
- Silhouette is highest at K=2 (0.41) and falls to 0.37 at K=3. The elbow shows a clear drop of about 30% at K=3, then smaller drops of 15 to 18% at K=4 to 6.

Segment	Learners	Median CTC	Mean CTC	Median experience
Lower-pay / early career	94,712 (61.7%)	₹6.8L	₹7.5L	4 yrs
Higher-pay / experienced	58,731 (38.3%)	₹19.9L	₹22.9L	9 yrs

- Median pay is **2.9× higher** in the experienced segment. Engineering Leadership is 8.8% of that segment against 0.8% of the early-career one.

- Adding role one-hot columns drops silhouette from 0.405 to 0.271, so pay and experience define the segments, not job family.
- Hierarchical clustering gives a similar split (medians ₹6L and ₹17L; silhouette 0.376), and agreement with K-Means is **ARI = 0.654**. The segments are reasonably robust but share a fuzzy boundary, since pay is a continuum.

Business recommendations

1. **Run two track families.**
    - Evidence: 61.7% of learners are early-career, with a ₹6.8L median at 4 years.
    - Action: build placement and interview-prep programmes for them, with affordable or outcome-linked pricing.
    - For the experienced 38.3% (₹19.9L median), build leadership and system-design programmes. Leadership roles are 11 times more concentrated there.
2. **Launch a salary-benchmark tool with a fallback chain.**
    - Evidence: Designation covers only 38.9% of learners, Class 62.3% and Tier 78%.
    - Action: show median and P25/P75 bands at Designation level when there are enough peers, then fall back to Class and Tier, labelling the comparison level clearly.
3. **Target the "paid below peers" group.**
    - Evidence: 30,354 learners (20.7%) are below peer mean on both Designation and Tier, and Designation-3 learners have a median of ₹6.4L against ₹15L for Designation-1.
    - Action: offer upskilling, negotiation coaching and job-switch support. Describe them as "paid below peers", not "underpaid", since the flag is mean-based.
4. **Use the pay-progression story in marketing.**

    - Evidence: median pay is 2.9× higher at about 9 years than at about 4.
    - Action: present it as an indicative range from cross-sectional data, not a guaranteed outcome.
5. **Prioritise employer partnerships by sample size.**
    - Evidence: the top median companies are mostly based on 20 to 41 records.
    - Action: start with dgzo (n=148) and validate the others with more data before building outreach around them.
6. **Position data/ML as a moderate premium.**
    - Evidence: a median of ₹12L against ₹10L overall, from about 3% of records.
    - Action: price it as a modest premium and test demand before a large investment. Backend and Full-Stack remain the core volume.
7. **Fix data capture.**
    - Evidence: 25.5% missing roles, 1,016 free-text role values, 4.66% out-of-range CTC and 25% repeated rows.
    - Action: use a role dropdown, add CTC range checks and unit labels (annual, in ₹), validate joining year, and timestamp or de-duplicate profile updates.

Limitations

- The data is self-reported and cross-sectional, with no inflation adjustment across update years.
- Outlier removal caps the top end at ₹92 lakh, so true top earners are excluded.
- Company names are anonymised, and role is missing for a quarter of learners.
- Silhouette of about 0.4 means the segments are useful summaries of a continuum, not sharply separated groups.